

10.3 Justifying Measures of Center and Variation

Lesson Introduction

 Benchmarks	MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation. MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.		 Learning Target	I can justify whether or not a measure of center or variation appropriately summarizes data.
 Benchmark Coherence	Building On MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		Working Toward MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate. MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.	
 Essential Questions	- How do you determine which measures of center best describe a data set? - How do you determine which measures of variation best describe a data set?			
 Lesson Narrative	In this lesson, students build on their understanding of the effect of outliers from Lesson 2. Students begin by analyzing what information can be determined from different data displays. Students then draw conclusions about how outliers or the shape of a distribution impact the mean, median, range, and IQR and apply this knowledge throughout the rest of the lesson to determine which measures are best to describe data sets given the context of the data presented.			
 Component Summary	10.3.1 Warm-Up (~5 min.)	Notice/Wonder about data in different formats (<i>SE p. 116</i>)	10.3.3 Guided Instruction (10 min.)	Intro to stem-and-leaf plots (<i>SE p. 118</i>)
	10.3.2 Collaboration (10-12 min.)	Review measures of center, variation, and types of data displays (<i>SE p. 117</i>)	10.3.4 Collaboration (15-18 min.)	Describe and interpret measures of center and spread from data displayed in histograms and stem-and-leaf plots (<i>SE p. 119</i>)
	10.3.5 Reflection (~5 min.)	Self-assessment of confidence in the day's learning targets (<i>SE p. 122</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Stem-and-Leaf Plot <u>Additional Terms:</u> - Mean - Range - Median - Interquartile range		Calculators	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions - Students may struggle to estimate the mean and median and determine which will be larger based on the shape of a distribution. - Students may not understand which measure of center or variation best describes a given data set. - Students may struggle to connect context to the measures of center and variation.	Things to Consider Consider encouraging students to refer back to the graphs and data displays to help visualize the data. Considering the shape of the distribution is helpful when determining the most appropriate measures of center and spread.
	Literacy and Language Routines Think-Pair-Share 10.3.4 Collaboration provides an opportunity to engage students in a Think-Pair-Share routine as they analyze student responses to data. Have each student think individually about the problem, then discuss it with their partner. As students talk, circulate to find 1-2 responses to share with the class that will advance student thinking and learning.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts This lesson provides a multitude of opportunities for students to engage in discourse that connects mathematics to real-world contexts. Students consider how the context of the data set impacts which measures of center and variation are most appropriate to describe the data.
	 Differentiate Instruction Throughout this lesson, visual entry points are provided by displaying data using various displays such as line plots and histograms. Auditory entry points can be provided through partner and class discussions.	
Lesson Notes:		

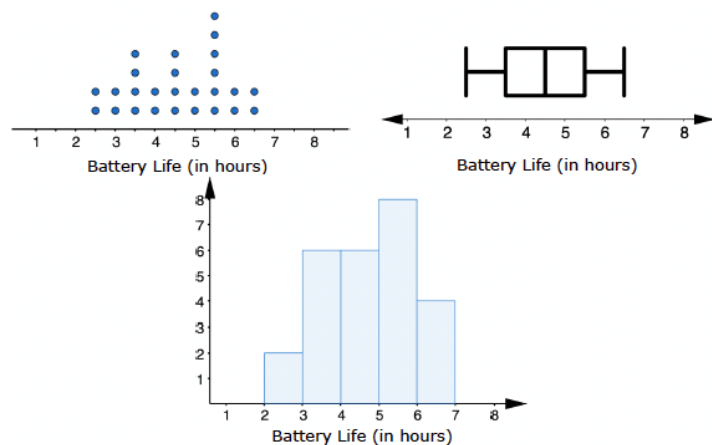
10.3.1 Warm-Up: Notice and Wonder

Component Overview: In this Warm-Up, students will discuss what they notice and wonder about the data shown in multiple formats.



10.3.1 Warm-Up: Notice and Wonder

The dot plot, histogram, and box plot summarize the hours of battery life for 26 cell phones that are constantly streaming videos.



Notice and Wonder

Engage students in a Notice and Wonder routine.

Encourage students to record similarities and differences they notice between the line plot, histogram, and box plot. Ask students what this makes them wonder about the data set or the data displays.

1. What do you notice?

Answers will vary. The dot plot and the histogram look the most similar. The median of the box plot looks to be very close to the middle.

2. What do you wonder?

Answers will vary. I wonder if all the batteries are the same. I wonder if the age of the phone affects the battery life.

10.3.2 Collaboration: Comparing Data Displays

Component Overview: Students revisit the different data displays for numerical data previously learned as well as the concept of an outlier.



10.3.2 Collaboration: Comparing Data Displays

Suppose an office tracks the number of phone calls received on business days, Monday through Friday, over two weeks. The data set is shown below.

183	165	58	152	179
152	153	135	124	112

- Discuss with your partner what you notice about the data set.

Answers may vary. There are two 152's. Most numbers are in the hundreds. Fifty-eight looks out of place.

- Match the measures of center and variation with their calculations for the data set above.

A. Mean	B. Median	C. Range	D. IQR
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B	$\frac{152+152}{2} = 152$
D	$165 - 124 = 41$
A	$\frac{183 + 165 + 58 + 152 + 179 + 152 + 153 + 135 + 124 + 112}{10} = 141.3$
C	$183 - 58 = 125$



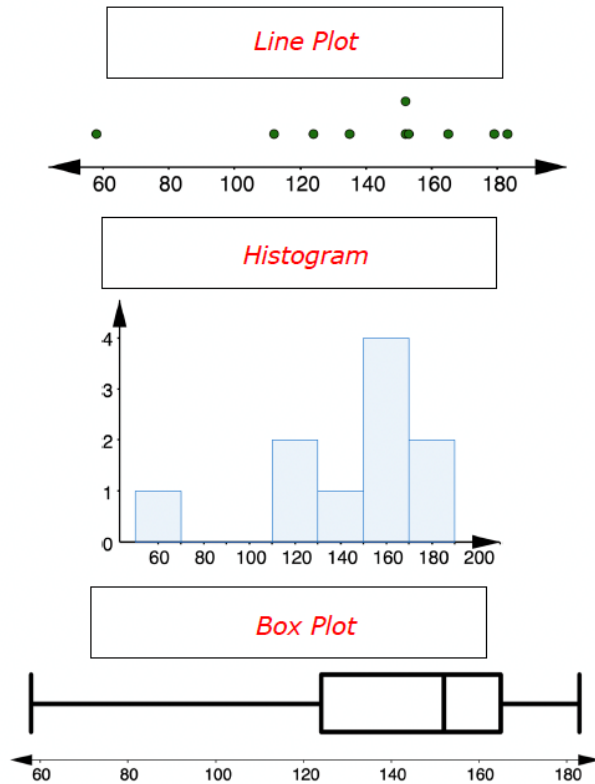
Notice and Wonder

To position students to begin to conceptually consider the concept of outliers, which will be reviewed at the end of this component, engage students in a Notice and Wonder routine. Ask questions like:

- What do you notice about the data set?
- What do the numbers have in common?
- What is different?
- What does this make you wonder?

10.3.2 Collaboration: Comparing Data Displays (continued)

3. A **histogram**, **line plot**, and **box plot** representing the data are shown.



Students represented data using these graphs in Unit 9. Activate prior knowledge through questioning:

- What are the components of a line plot? Histogram? Box plot?
- What do you remember about each of these data displays?
- Which measures of center and variation can be determined from each display?

- a. Label the display types histogram, box plot, or line plot by filling in the box above each display.

10.3.2 Collaboration: Comparing Data Displays (continued)

- b. Complete the table by identifying which measures of center and variation can be determined from each display.

	Mean	Median	Range	IQR
Histogram	☐	☐	☐	☐
Line Plot	☐	☐	☐	☐
Box Plot	☐	☐	☐	☐

4. Describe any outliers in the data set.

58 is an outlier because it is far away from the rest of the values.



5. Explain which of the displays you used to determine whether or not there was an outlier in the data set.

Answers will vary. I used the line plot because I could see that this data point was far away from the other points in the data set. I used the box plot to see that the lowest data point was far from the median.

Students do not need to know how to determine whether a number is an outlier in the Grade 6 Accelerated course, so the values used as potential outliers have already been confirmed in these lessons.

Consider explaining to students that there is a calculation to determine if a value in a data set is actually an outlier. They should not just assume any number larger or smaller than the others in a data set is automatically an outlier, though the examples they'll see in this unit are.

For the teacher's reference, a number is an outlier if it is $1.5 \times \text{IQR}$ above $Q3$ or below $Q1$ in a set of data.

10.3.3 Guided Instruction: Stem-and-Leaf Plots

Component Overview: Introduce students to stem-and-leaf plots, another way to organize and display numerical data.



10.3.3 Guided Instruction: Stem-and-Leaf Plots

Recall the data collected on the number of phone calls an office received on business days over two weeks.

183	165	58	152	179
152	153	135	124	112

In addition to displaying the data graphically with a histogram, line plot, or box plot, the data can also be displayed in a **stem-and-leaf** plot.



A **stem-and-leaf plot** is a table that organizes data by place value to compare data frequencies.

1. What do you notice about the data in the stem-and-leaf plot?

Answers will vary. There is a large gap in the data. There is a number that repeats, 152.

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58



Notice and Wonder

Review the definition of stem-and-leaf plot with students, then use the plot in question 1 to draw students' attention to the features of stem-and-leaf plots. Engage students in a Notice and Wonder routine and circulate as they discuss in groups or pairs.

Students may connect the stem-and-leaf plot to a vertical number line where each ten, in this case, is marked, but there are only values in the second column when there is an actual data value from the set that falls within that interval of 10. Accounting for each interval of 10 within the range allows the display to show the spread of the data.



Stem-and-leaf plots are a useful tool. Students can visually see the large gap in the data, meaning outliers can be more easily identified. Consider creating an anchor chart with students that identifies the parts of stem-and-leaf plots for students to reference throughout this unit.

10.3.3 Guided Instruction: Stem-and-Leaf Plots (continued)

2. Compare the stem-and-leaf plot with the line plot from the collaboration.

What similarities do you notice between the displays?	What differences do you notice between the displays?
<i>Answers will vary. Both displays show a gap in the data.</i>	<i>Answers will vary. The stem-and-leaf plot shows exact values. The line plot's scale makes it hard to determine what the value of each data point is from the display.</i>

3. What value does the entry 13|5 represent? **135**

4. Which measure(s) of center can be determined from the stem-and-leaf plot?

- ☐ mean
- ☐ median
- ☒ both the mean and median

5. Which measure(s) of variation can be determined from the stem-and-leaf plot?

- ☐ range
- ☐ interquartile range
- ☒ both the range and the interquartile range



Display both plots for students to see. Have students work with a partner to discuss the similarities and differences between the displays. As students compare the displays, circulate and ask questions like:

- What do you notice about the values on each display? What does this mean for determining the measures of center and variation?
- What do you notice about the scale of the line plot? Why is this important?



What does it mean when there are no values in the leaf column for some of the stems?

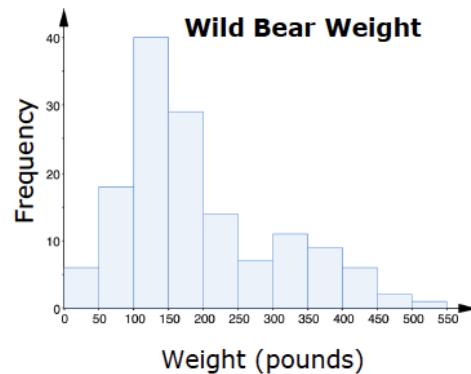
10.3.4 Collaboration: Center and Variation in Context

Component Overview: Students work with a partner to analyze histograms and stem-and-leaf plots. Students use the data displays to answer questions about the data and draw conclusions about the measures of center and measures of variation that best describe each distribution in context.



10.3.4 Collaboration: Center and Variation in Context

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- a. Complete the statements.

Based on the data, it appears wild bears typically

weigh between
☐ 50 and 150
☒ 100 and 200
☐ 150 and 250
 pounds. A good

estimate for the typical weight for wild bears is 175

pounds.



Students may struggle to understand what a “typical weight” is in this context. Consider asking guiding questions like:

- Between what numbers do most of the weights fall in the data set?
- What does typical mean in this context? Is it referring to the center or variation of the data set?
- If the most weights are between 100-150 and 150-200, how can you describe the typical weights of the wild bears?

10.3.4 Collaboration: Center and Variation in Context (continued)

- b. Compare your estimate with your partner. Discuss whether it is possible for both of your estimates to be reasonable, even if they are not the same.

- c. Discuss with your partner whether your estimate from Part A is closest to the median or the mean. Summarize your discussion.

Answers will vary. The estimate would be closest to the median. The few bears in the data set that weigh more will cause the mean to be higher than the median.

- d. Angelica and Raymond were asked to describe the variation in weight of the wild bears.

Angelica said, "The bear weights range from less than 50 to over 500 lbs, so the range best describes the spread of the weights of the wild bears."

Raymond said, "It looks like most bears in the sample weigh between 100 and 300 lbs, with few weighing less and few weighing more. The small weights could be because there were babies in the sample. So, the interquartile range is the best measure of spread to describe this data."

Explain whether you agree with Angelica or Raymond. Justify your reasoning.

Answers will vary. Raymond's explanation best describes the spread. He justifies his choice with the understanding that the small weights are because the bears were babies and the frequency for the weights between 100 and 300 is much higher than at heavier weights.



Auditory entry points are built in throughout this component through partner discussion. Consider providing additional visual entry points through the use of diagrams. Encourage students to use colored pencils to mark the median and mean on the histogram to help visualize their estimations and response.

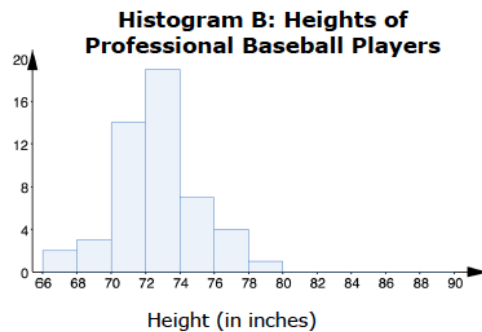
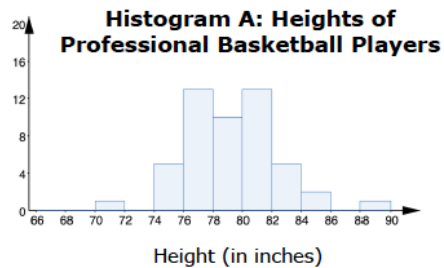


Think-Pair-Share

Consider implementing a Think-Pair-Share routine for question 1d. Have each student think individually about the problem, then discuss it with their partner. As students talk, circulate to find 1-2 responses to share with the class that will advance student thinking and learning. Listen for students who can justify why the IQR makes the most sense based on the shape of the distribution.

10.3.4 Collaboration: Center and Variation in Context (continued)

2. The two histograms, labeled A and B, show height distributions of 50 male professional basketball players and 50 male professional baseball players.



Work with your partner to complete each statement.

- a. In ☐ Histogram A, ☒ Histogram B, the average height is around 72 inches. The values of the mean and median are similar.
- b. To explain typical height, the variation in ☒ Histogram A ☐ Histogram B is best described by the interquartile range because the range may be misleading due to outliers. The values of the mean and median are similar.



Students may easily guess the answers to question 2. However, connect it to the next activity by asking questions like:

- What values appear to be outliers?
- What would the mean and median be of each histogram?
- Is the mode easiest to find when given a histogram?
- On average are basketball players or baseball players taller?

10.3.4 Collaboration: Center and Variation in Context (continued)

3. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer displayed the earthworms' lengths, in millimeters (mm), in a stem-and-leaf plot.

Stem	Leaf
0	6
1	1, 8, 9
2	0, 3, 3, 5, 5, 6, 7, 7, 8, 9
3	2, 3
4	1, 2, 8
5	2, 4, 9
6	0
7	7
8	
9	3

Key: $2|8 = 28$

Use the display to answer the following questions.

- Half of the earthworms sampled were less than **28** mm long.
- The farmer reported that the average length is 35.92 mm. Why might this measure of center be misleading about the typical length of the earthworms in this sample?
Answers will vary. This could be misleading because of the vast range of data.
- Compare your answers to Parts A and B with your partner, and if necessary, resolve any differences.



If students are struggling to answer question 3b, provide additional scaffolding through intentional questions like:

- What do you notice about the spread of the data? Why is this important when considering measure of center?
- Based on what you know so far about measures of center, how can outliers impact the mean, or average?
- Is there one measure of center that would be more accurate to describe the data? Why?

10.3.4 Collaboration: Center and Variation in Context (continued)

- d. The farmer reported that the lengths varied between 6 mm and 93 mm. The farmer's assistant said that most earthworms had lengths between 23 and 50 mm.

Work with your partner to complete the statements.

The farmer is using the ☐ IQR
☒ range to summarize

the variation in worm length, while the farmer's

assistant is using the ☒ IQR.
☐ range.

If a researcher is interested in how long earthworms are, the ☒ IQR
☐ range is an appropriate measure of variation for the farmer to report

because *Answers will vary. IQR is not affected by outliers.*



While students are working, consider any observational data based on student responses and discussions when completing 10.3.4 Collaboration. Use the observations to inform what differentiation or small-group instruction can look like in future lessons to ensure students are able to correctly determine which measures of center and variation to use to describe a data set as well as how to interpret those measures in context.

10.3.5 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection to indicate their current understanding of the lesson's learning targets.



10.3.5 Wrap-Up: Reflection

Draw an **X** on the line to indicate where your current learning is in relation to the following learning targets.

1. I can justify whether or not a measure of center appropriately summarizes data.

Answers will vary.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

2. I can justify whether or not a measure of variation appropriately summarizes data.

Answers will vary.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.

